

Experiment # 13

NETWORK ADDRESS TRANSLATION

Objective:

This Lab explains how to implement NAT protocol.

Software Tools:

- Cisco Packet Tracer

Theory:

NETWORK ADDRESS TRANSLATION

Introduction:

Network address translation (NAT) is a method of remapping one IP **address** space into another by modifying **network address** information in IP header of packets while they are in transit across a traffic routing device.

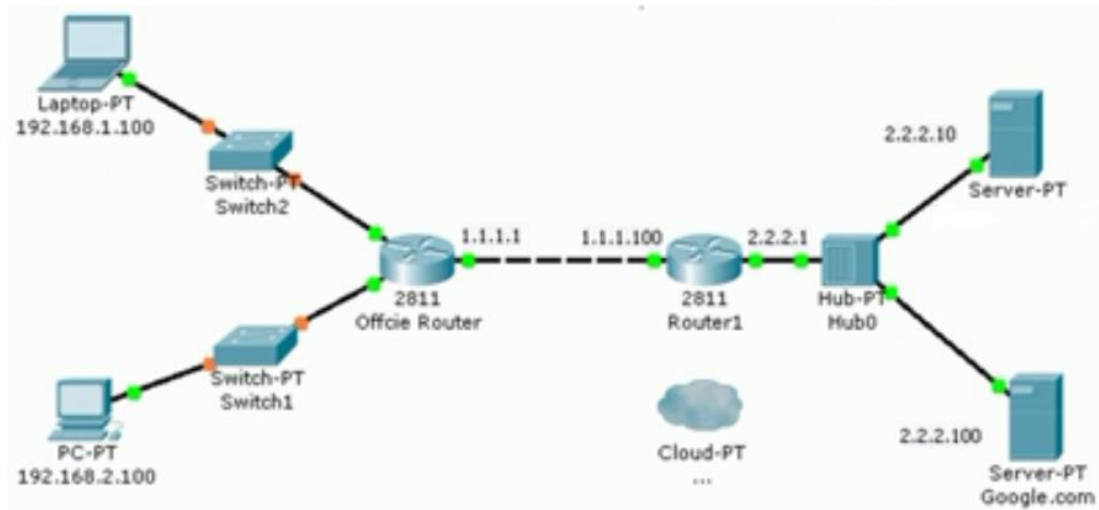
It enables private IP networks that use unregistered IP addresses to connect to the Internet. **NAT** operates on a router, usually connecting two networks together, and translates the private (not globally unique) addresses in the internal network into legal addresses, before packets are forwarded to another network.

In short, NAT allows only a single globally unique IPv4 address to represent an entire network to the outside world.

Three types of NAT commonly in use are

- 1) Static NAT
- 2) Dynamic NAT
- 3) Port Address Translation (PAT/NAT Overload)

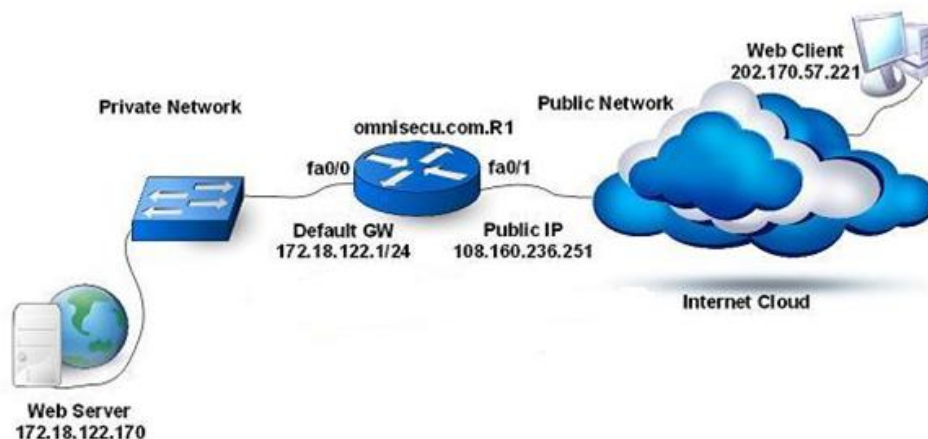
Lab setup:



How to configure static NAT in a Cisco Router

Static NAT (Network Address Translation) is one-to-one mapping of a private IP address to a public IP address. Static NAT (Network Address Translation) is useful when a network device inside a private network needs to be accessible from internet.

For example, we have a web server inside our network, which is configured with a private IP address as shown below.



We want to connect our web server with outside world, so that general public can visit our web pages. Our webserver is configured with IP address 172.18.122.170, which is a private IP address. The webserver is not directly accessible from the public internet because all the traffic originating from or destined to private IP addresses are dropped by public internet routers. Here static NAT can be used to allow forwarding traffic from

the internet to a server inside our network with a private IP address. Generally, this technology is known as port forwarding.

When web clients try to make a connection to our webserver, they connect using a public IP address configured on our NAT (Network Address Translation) router (HTTP - TCP port number 80). The connection is then translated and send to the webserver. The HTTP response from the webserver is send back to the HTTP client.

The static NAT configuration command syntax for a Cisco Router is as below.

```
uet.R1(config)#ip nat inside source static tcp <inside_local_ip_address> <inside_local_port>
<inside_global_ip_address> <inside_global_port>
```

After configuring static NAT using above command, you have to identify which is the inside interface (facing the private network) and which is the outside interface (facing the public internet) using *ip nat inside* and *ip nat outside* from interface configuration mode.

The configuration commands are as below.

```
uet.R1>enable
uet.R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.

uet.R1(config)#ip nat inside source static tcp 172.18.122.170 80 108.160.236.251 80

uet.R1(config)#interface fa0/1
uet.R1(config-if)#ip nat outside
uet.R1(config-if)#interface fa0/0
uet.R1(config-if)#ip nat inside
uet.R1(config-if)#exit
uet.R1(config)#exit
```

Interface fa0/1 is the interface which is facing public internet and fa0/0 is the interface which is facing private inside network.

[How to configure Dynamic NAT in a Cisco Router](#)

Dynamic NAT is another NAT (Network Address Translation) technology which allows the address translation of a private IP address to a pool of public IP addresses configured on the NAT router. Dynamic NAT is mostly used when inside computers configured with private IP addresses needs to access outside public internet.

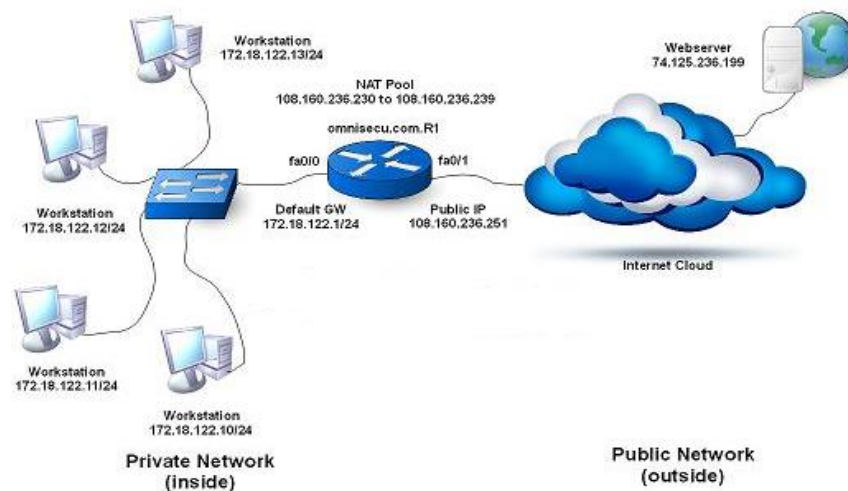
The main difference between Static NAT and Dynamic NAT is that Static NAT allows a remote host to connect to an inside private IP addresses configured computer and Dynamic NAT allows a group of private IP addresses to connect to public internet, using the public IP address pool (a range of public IP addresses). The configured public IP address pool (a range of public IP addresses) typically has fewer addresses than the inside private IP addresses.

Dynamic NAT is typically used for providing internet access to a private network. But the problem with Dynamic NAT is that the number of public IP addresses in the NAT pool may not be sufficient for mapping large number of computers configured with private IP addresses.

To configure Dynamic NAT on a Cisco router, first you need to create an access list to identify the group of private inside IPv4 addresses, which are allowed for NAT translation. That can be done by creating a standard IP access list.

Then you have to create a pool of public IP addresses (which your ISP (internet service provider) has allocated to you).

After creating an IP access list to identify the traffic and a pool of public IP addresses which your ISP (internet service provider) has allocated to you, you must configure NAT using “ip nat” command. Finally you must specify which is inside interface and which is outside interface.



The configuration commands to configure Dynamic NAT is shown below.

```
uet.R1#configure terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
uet.R1(config)#access-list 10 permit 172.18.122.0 0.0.0.255
uet.R1(config)#ip nat pool uetPool 108.160.236.230 108.160.236.239 netmask 255.255.255.0
uet.R1(config)#ip nat inside source list 10 pool uetPool
uet.R1(config)#interface fa0/0
uet.R1(config-if)#ip nat inside
uet.R1(config-if)#exit
uet.R1(config)#int fa0/1
uet.R1(config-if)#ip nat outside
uet.R1(config-if)#exit
uet.R1(config)#exit
uet.R1#
```

How to configure PAT (NAT Overload) in a Cisco Router

The configuration commands to configure PAT (Port Address Translation or NAT overload) is shown below.

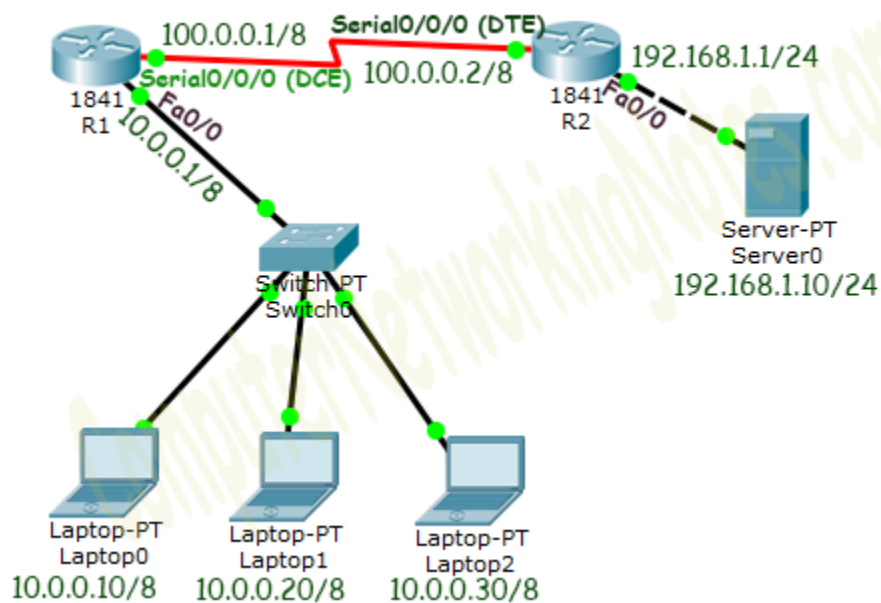
```
uet.R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

uet.R1(config)#access-list 10 permit 172.18.122.0 0.0.0.255
uet.R1(config)#ip nat pool uetPool 108.160.236.251 108.160.236.251 netmask 255.255.255.0

uet.R1(config)#ip nat inside source list 10 pool uetPool overload
uet.R1(config)#interface fa0/0
uet.R1(config-if)#ip nat inside
uet.R1(config-if)#exit
uet.R1(config)#int fa0/1
uet.R1(config-if)#ip nat outside
uet.R1(config-if)#exit
uet.R1(config)#exit
```

LAB TASK

Simulate below Topology:



Device / Interface	IP Address	Connected With
Laptop0	10.0.0.10/8	Fa0/0 of R0
Laptop1	10.0.0.20/8	Fa0/0 of R0
Laptop2	10.0.0.30/8	Fa0/0 of R0
Server0	192.168.1.10/24	Fa0/0 of R1
Serial 0/0/0 of R1	100.0.0.1/8	Serial 0/0/0 of R2
Serial 0/0/0 of R2	100.0.0.2/8	Serial 0/0/0 of R2